Graphical Abstract

This is a specimen a_b title

J.K. Krishnan, Han Thane, William J. Hansen, T. Rafeeq

Highlights

This is a specimen a_b title

J.K. Krishnan, Han Thane, William J. Hansen, T. Rafeeq

- Research highlights item 1
- Research highlights item 2
- Research highlights item 3

This is a specimen a_b title[⋆,⋆⋆]

Sir J.K. Krishnan[a,c,*,1] (Researcher), Han Thane[b,d], William J. Hansen Jr[b,c,2] (Co-ordinator) and T. Rafeeq[a,c,**,1,3]

[a]Department of Physics, J.K. Institute of Science, Jawahar Nagar, Trivandrum, 695013, Kerala, India
[b]World Scientific University, Street 29, 1011 NX Amsterdam, The Netherlands
[c]University of Intelligent Studies, Street 15, Jabaldesh, 825001, Orissa, India

ARTICLE INFO

Keywords:
quadrupole exciton
polariton
WGM
BEC

ABSTRACT

This template helps you to create a properly formatted LaTeX manuscript.
`\beginabstract ... \endabstract` and `\begin{keyword} ... \end{keyword}` which contain the abstract and keywords respectively.
Each keyword shall be separated by a `\sep` command.

1. Introduction

The Elsevier cas-dc class is based on the standard article class and supports almost all of the functionality of that class. In addition, it features commands and options to format the

- document style
- baselineskip
- front matter
- keywords and MSC codes
- theorems, definitions and proofs
- lables of enumerations
- citation style and labeling.

This class depends on the following packages for its proper functioning:

1. natbib.sty for citation processing;
2. geometry.sty for margin settings;
3. fleqn.clo for left aligned equations;
4. graphicx.sty for graphics inclusion;
5. hyperref.sty optional packages if hyperlinking is required in the document;

All the above packages are part of any standard LaTeX installation. Therefore, the users need not be bothered about downloading any extra packages.

2. Numerical methodology

2.1. Overall numerical framework

A two-dimensional steady aero–hydraulic framework was developed to evaluate the aerodynamic behaviour of airfoils containing passive internal channels. The external flow is modelled using an incompressible, inviscid and irrotational source–vortex panel method, while the pressure-driven flow through each internal passage is described using the Hagen–Poiseuille and Darcy–Weisbach relations for circular channels. The two models are coupled through the pressure and surface-normal velocity at the channel openings.

The external aerodynamic solution determines the pressure difference between the two openings of each channel. This pressure difference drives the internal flow, which subsequently modifies the aerodynamic boundary condition through a surface-normal transpiration velocity. The external pressure field and internal channel flow are therefore determined as a coupled aero–hydraulic system.

The calculation is initialised using an impermeable-airfoil solution, for which the surface-normal transpiration velocity is set to zero. The resulting surface pressures provide the initial pressure differences across the internal passages and, consequently, the initial estimates of the channel flow rates. The aerodynamic and hydraulic solutions are then coupled until a consistent pressure and channel-flow solution is obtained. Converged solutions are retained for aerodynamic and internal-flow analysis, whereas non-converged cases are reported as `NaN`.

The external aerodynamic formulation assumes steady, incompressible, inviscid and irrotational flow. Consequently,

---

⋆This document is the results of the research project funded by the National Science Foundation.

⋆⋆The second title footnote which is a longer text matter to fill through the whole text width and overflow into another line in the footnotes area of the first page.

This note has no numbers. In this work we demonstrate a_b the formation Y_1 of a new type of polariton on the interface between a cuprous oxide slab and a polystyrene micro-sphere placed on the slab.

*Corresponding author
**Principal corresponding author
jkk@example.in (J.K. Krishnan); wjh@example.org (W. J. Hansen); t.rafeeq@example.in (T. Rafeeq)
www.jkkrishnan.in (J.K. Krishnan); https://www.university.org (W. J. Hansen); www.campus.in (T. Rafeeq)
ORCID(s): 0000-0001-0000-0000 (J.K. Krishnan)

[1]This is the first author footnote, but is common to third author as well.
[2]Another author footnote, this is a very long footnote and it should be a really long footnote. But this footnote is not yet sufficiently long enough to make two lines of footnote text.

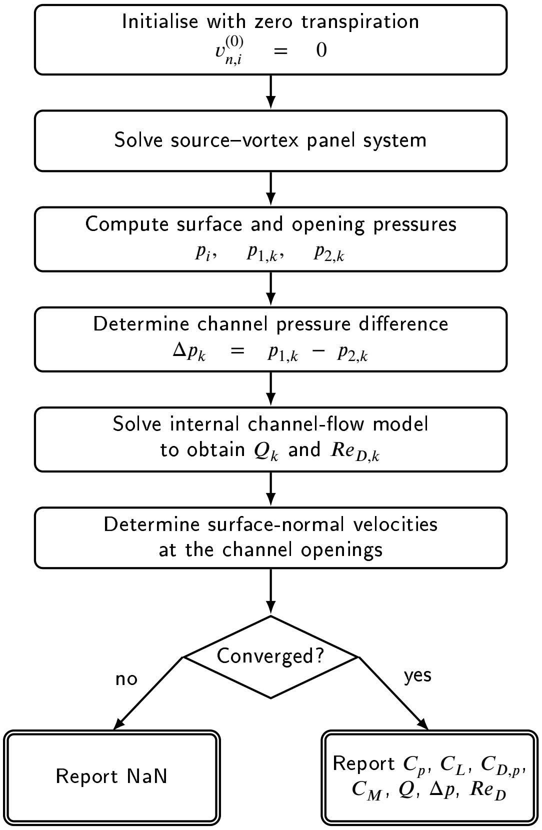
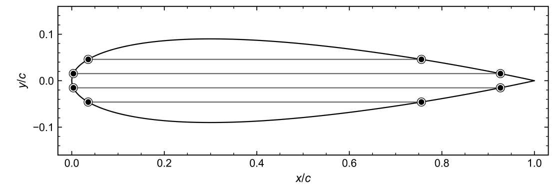
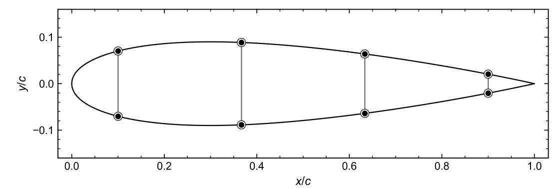


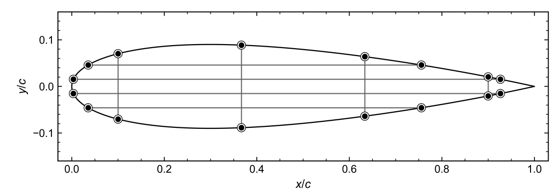
Figure 1: Coupled aero–hydraulic solution procedure.



(a) Horizontal model (M1)



(b) Vertical model (M2)



(c) Combined independent model (M3)

Figure 2: Porous-airfoil configurations considered in the present study: (a) horizontal model (M1), (b) vertical model (M2), and (c) combined independent model (M3).

boundary-layer development, skin-friction drag, flow separation and stall are not resolved. The predicted drag coefficient therefore represents only the pressure-drag contribution obtained from the potential-flow solution.

The numerical procedure is summarised in Fig. 1.

2.2. Airfoil geometry and porous configurations

The baseline geometry is a NACA 0018 airfoil with chord c. The NACA four-digit coordinates are generated analytically and discretised using cosine spacing along the chord, providing increased panel density near the leading and trailing edges.

Each aerodynamic panel i is characterised by its length S_i, control point (x_i, y_i), unit tangent vector

$$\boldsymbol{t}_i = (t_{x,i}, t_{y,i}), \tag{1}$$

and outward unit normal vector

$$\boldsymbol{n}_i = (n_{x,i}, n_{y,i}). \tag{2}$$

The porous structure is represented by passive circular channels connecting pairs of openings on the airfoil surface. No prescribed mass-flow rate or external pumping mechanism is imposed. Instead, the magnitude and direction of the internal flow are determined entirely by the aerodynamic pressure difference between the two surface openings.

Three porous configurations are considered:

1. **Horizontal model (M1):** four independent chordwise passages, with two channels located above and two below the chord line. Both openings of each passage are located on the same airfoil surface.
2. **Vertical model (M2):** four independent lower-to-upper passages distributed between $x/c = 0.1$ and $x/c = 0.9$.
3. **Combined independent model (M3):** the horizontal and vertical passages are superposed, resulting in eight internal passages. The passages remain hydraulically independent, and geometrical intersections do not represent connected internal junctions.

The three porous-airfoil configurations considered in the present study are illustrated in Fig. 2.

The positions of the surface openings are obtained by interpolation of the upper and lower airfoil surfaces rather than by restricting their locations to panel control points. This preserves the prescribed orientation and position of the internal passages independently of the aerodynamic panel resolution.

The channel geometry is checked before the coupled calculation to ensure that the openings remain within the permitted chordwise region, do not overlap, fit within the local airfoil thickness and satisfy the specified minimum separation.

Each passage k is characterised by a diameter D_k and a centreline length L_k. The channel length is determined from the geometry of the two corresponding surface openings and the internal passage connecting them.

2.3. External aerodynamic model

2.3.1. Potential-flow formulation

The external flow is assumed to be incompressible, inviscid and irrotational. The velocity field can therefore be expressed in terms of a scalar velocity potential,

$$\boldsymbol{V} = \nabla\Phi. \tag{3}$$

Application of mass conservation gives Laplace's equation,

$$\nabla^2\Phi = 0. \tag{4}$$

The numerical solution is constructed by superposing the uniform freestream with constant-strength source distributions over the airfoil surface and a single constant vortex strength distributed around the complete airfoil boundary. The source distributions represent the airfoil thickness, while the global vortex strength introduces the circulation required to generate lift.

2.3.2. Source–vortex panel formulation

For an airfoil discretised into N panels, the aerodynamic unknowns are the N source strengths

$$\lambda = \left[\lambda_1, \lambda_2, \dots, \lambda_N\right]^{\mathrm{T}} \tag{5}$$

and the global vortex strength γ.

The normal and tangential source influence coefficients are denoted by I_{ij} and J_{ij}, respectively, while K_{ij} and L_{ij} represent the corresponding vortex influence coefficients.

For an impermeable airfoil, the normal velocity at every control point is zero. To represent a porous opening, this condition is modified by prescribing a surface-normal transpiration velocity $v_{n,i}$,

$$\boldsymbol{V}_i \cdot \boldsymbol{n}_i = v_{n,i}. \tag{6}$$

The conventional impermeable boundary condition is recovered when

$$v_{n,i} = 0. \tag{7}$$

Following discretisation, the normal boundary condition for panel i becomes

$$\pi\lambda_i + \sum_{\substack{j=1 \\ j\neq i}}^{N} I_{ij}\lambda_j - \gamma\sum_{j=1}^{N} K_{ij} = -2\pi V_\infty \cos\beta_i + 2\pi v_{n,i}, \tag{8}$$

where β_i denotes the angle between the local panel normal and the freestream direction.

The trailing-edge Kutta condition is imposed by requiring the tangential velocities on the first and final panels to be equal in magnitude and opposite in direction,

$$V_{t,1} + V_{t,N} = 0. \tag{9}$$

The N normal boundary conditions together with the Kutta condition produce an $(N+1)\times(N+1)$ linear system,

$$\begin{bmatrix} \boldsymbol{A}_\lambda & \boldsymbol{a}_\gamma \\ \boldsymbol{a}_K^{\mathrm{T}} & a_{K,\gamma} \end{bmatrix} \begin{bmatrix} \lambda \\ \gamma \end{bmatrix} = \begin{bmatrix} \boldsymbol{b}(\boldsymbol{v}_n) \\ b_K \end{bmatrix}. \tag{10}$$

The influence matrix depends only on the discretised airfoil geometry. It is therefore assembled and LU-factorised once for each geometry, while the right-hand side is updated according to the freestream condition and surface-normal transpiration velocity.

2.3.3. Surface velocity and pressure

Following solution of the panel system, the tangential velocity on panel i is evaluated as

$$V_{t,i} = V_\infty \sin\beta_i + \frac{1}{2\pi}\sum_{j=1}^{N} \lambda_j J_{ij} + \frac{\gamma}{2} - \frac{\gamma}{2\pi}\sum_{j=1}^{N} L_{ij}. \tag{11}$$

At a porous opening, the local surface velocity contains both tangential and normal components. The velocity magnitude is therefore

$$V_i^2 = V_{t,i}^2 + v_{n,i}^2. \tag{12}$$

The pressure coefficient is obtained from the steady incompressible Bernoulli equation,

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + v_{n,i}^2}{V_\infty^2}. \tag{13}$$

The corresponding absolute pressure is

$$p_i = p_\infty + \frac{1}{2}\rho_\infty V_\infty^2 C_{p,i}. \tag{14}$$

2.4. Internal channel-flow model

The internal flow through each porous passage is modelled as pressure-driven flow through a circular channel. For channel k with diameter D_k, the cross-sectional area is

$$A_k = \frac{\pi D_k^2}{4}. \tag{15}$$

The mean internal velocity is related to the volumetric flow rate by

$$\overline{U}_k = \frac{Q_k}{A_k}, \tag{16}$$

where Q_k [$\mathrm{m^3\,s^{-1}}$] is the volumetric flow rate through the channel.

The internal flow is driven by the pressure difference between the two surface openings,

$$\Delta p_k = p_{1,k} - p_{2,k}. \tag{17}$$

The sign of Δp_k determines the direction of the internal flow. The corresponding channel Reynolds number is

$$Re_{D,k} = \frac{\rho_\infty |\overline{U}_k| D_k}{\mu_\infty}. \tag{18}$$

For fully developed laminar flow through a circular channel, the pressure-driven flow is described by the Hagen–Poiseuille equation,

$$Q_k = \frac{\pi D_k^4}{128\mu_\infty L_k}\Delta p_k. \tag{19}$$

This formulation naturally accounts for the direction of flow through the sign of Δp_k.

The pressure loss can equivalently be expressed using the Darcy–Weisbach relation,

$$|\Delta p_k| = f_k \frac{L_k}{D_k}\frac{\rho_\infty \overline{U}_k^2}{2}, \tag{20}$$

where f_k is the Darcy friction factor.

For fully developed laminar flow through a circular pipe,

$$f_{\mathrm{lam},k} = \frac{64}{Re_{D,k}}. \tag{21}$$

The Hagen–Poiseuille and Darcy–Weisbach formulations are therefore equivalent for fully developed laminar flow through the circular channels considered here.

2.5. Aero–hydraulic coupling

The aerodynamic and internal-flow models are coupled through the pressure and surface-normal velocity at the channel openings.

The pressure at each channel opening is obtained from the aerodynamic surface-pressure distribution. When an opening intersects multiple panels, its pressure is evaluated using a length-weighted average over the corresponding surface footprint. For opening m of channel k,

$$p_{m,k} = \frac{\sum_{i\in\mathcal{P}_{m,k}} p_i \ell_i}{\sum_{i\in\mathcal{P}_{m,k}} \ell_i}, \qquad m = 1, 2, \tag{22}$$

where $\mathcal{P}_{m,k}$ denotes the set of panels intersected by the opening and ℓ_i is the corresponding overlap length.

The two opening pressures provide the channel pressure difference,

$$\Delta p_k = p_{1,k} - p_{2,k}, \tag{23}$$

which is supplied to the internal-flow model to determine Q_k.

The mean velocity through each circular opening is

$$U_{\mathrm{open},k} = \frac{Q_k}{A_k}. \tag{24}$$

This velocity is imposed on the external aerodynamic model as a surface-normal transpiration velocity. With the outward panel normal defined as positive, flow entering the airfoil is assigned a negative normal velocity, whereas flow leaving the airfoil is assigned a positive normal velocity,

$$v_{n,k}^{\mathrm{in}} = -\frac{Q_k}{A_k}, \qquad v_{n,k}^{\mathrm{out}} = +\frac{Q_k}{A_k}. \tag{25}$$

When an opening intersects multiple aerodynamic panels, the prescribed normal velocity is applied over the corresponding opening footprint so that the transpiration boundary condition is represented consistently by the discretised airfoil surface.

The updated surface-normal velocity modifies the panel-method boundary condition and therefore changes the external pressure distribution. The new opening pressures are subsequently supplied to the internal-flow model, and the aerodynamic and hydraulic solutions are coupled until the channel flow used to impose the opening velocity is consistent with that predicted from the resulting pressure difference.

A case is classified as converged when the prescribed coupling criterion is satisfied for all channels. Cases that do not satisfy the convergence requirement are reported as `NaN`.

2.6. Aerodynamic coefficients

The aerodynamic forces are obtained by integrating the pressure contribution over the discretised airfoil surface. The contribution of panel i to the global Cartesian force coefficients is

$$\mathrm{d}C_{x,i} = -C_{p,i} n_{x,i} \frac{S_i}{c}, \tag{26}$$

$$\mathrm{d}C_{y,i} = -C_{p,i} n_{y,i} \frac{S_i}{c}. \tag{27}$$

The total Cartesian force coefficients are

$$C_x = \sum_{i=1}^{N} \mathrm{d}C_{x,i}, \qquad C_y = \sum_{i=1}^{N} \mathrm{d}C_{y,i}. \tag{28}$$

Rotation into the freestream-aligned coordinate system gives the lift and pressure-drag coefficients,

$$C_L = C_y \cos\alpha - C_x \sin\alpha, \tag{29}$$
$$C_{D,p} = C_x \cos\alpha + C_y \sin\alpha. \tag{30}$$

The pitching-moment coefficient is evaluated about the quarter-chord reference location,

$$(x_{\mathrm{ref}}, y_{\mathrm{ref}}) = (0.25c, 0), \tag{31}$$

using

$$C_M = -\sum_{i=1}^{N} \frac{(x_i - x_{\mathrm{ref}})\mathrm{d}C_{y,i} - (y_i - y_{\mathrm{ref}})\mathrm{d}C_{x,i}}{c}. \tag{32}$$

For diagnostic comparison, an additional lift coefficient is calculated from the solved circulation using the Kutta–Joukowski relationship,

$$C_{L,\Gamma} = \frac{2\gamma \sum_{i=1}^{N} S_i}{V_\infty c}. \tag{33}$$

2.7. Numerical setup

The aerodynamic response of each porous configuration is evaluated relative to an impermeable reference airfoil using the same geometry, panel discretisation and operating conditions. An inviscid XFOIL solution may additionally be used for comparison of the impermeable-airfoil pressure distribution and angle-of-attack response.

Unless otherwise stated, the calculations are performed for a NACA 0018 airfoil with chord

$$c = 1\ \mathrm{m}, \tag{34}$$

discretised using $N = 1000$ panels.

The freestream fluid properties are

$$\rho_\infty = 1.225\ \mathrm{kg\,m^{-3}}, \tag{35}$$

$$\mu_\infty = 1.8 \times 10^{-5}\ \mathrm{Pa\,s}, \tag{36}$$

and

$$p_\infty = 101325\ \mathrm{Pa}. \tag{37}$$

A chord Reynolds number of

$$Re_c = 5.0 \times 10^5 \tag{38}$$

is prescribed.

The corresponding freestream velocity is calculated from

$$V_\infty = \frac{Re_c \mu_\infty}{\rho_\infty c}. \tag{39}$$

The principal fixed-angle analysis is performed at

$$\alpha = 4^\circ. \tag{40}$$

The angle-of-attack sweep extends from -5° to 15° in increments of 1°.

Circular channel diameters of

$$D = 8,\ 9,\ \text{and}\ 10\ \mathrm{mm} \tag{41}$$

are considered.

Each angle of attack is solved independently, and the solution from the preceding angle is not used to initialise the subsequent case.

3. Results and discussion

The aerodynamic response of the three porous-airfoil configurations was investigated over an angle-of-attack range of -5° to 15°. Three circular-channel diameters, $D = 8$, 9 and 10 mm, were considered for the horizontal (M1), vertical (M2) and combined horizontal–vertical (M3) configurations. The corresponding impermeable panel-method solution and the inviscid XFOIL prediction are included as reference cases.

The influence of the porous channels is first assessed using the integrated lift and pitching-moment coefficients. The resulting aerodynamic trends are then related to changes in the external pressure and velocity fields. The flow-field comparison is presented for $\alpha = 10^\circ$ and $D = 9$ mm.

3.1. Baseline aerodynamic validation

The impermeable source–vortex panel solution is compared with the corresponding inviscid XFOIL prediction in Figs. 3 and 4. The two approaches show similar angle-of-attack trends for both lift and pitching moment over the investigated range.

The agreement between the two impermeable-airfoil solutions provides a reference for evaluating the aerodynamic changes introduced by the internal channels. Small differences between the panel-method and XFOIL predictions are expected because the two methods employ different numerical formulations, although both represent inviscid external flow.

The impermeable panel solution is therefore used as the primary reference when evaluating the effect of the porous configurations, ensuring that the porous and solid cases are compared using the same panel discretisation and aerodynamic formulation.

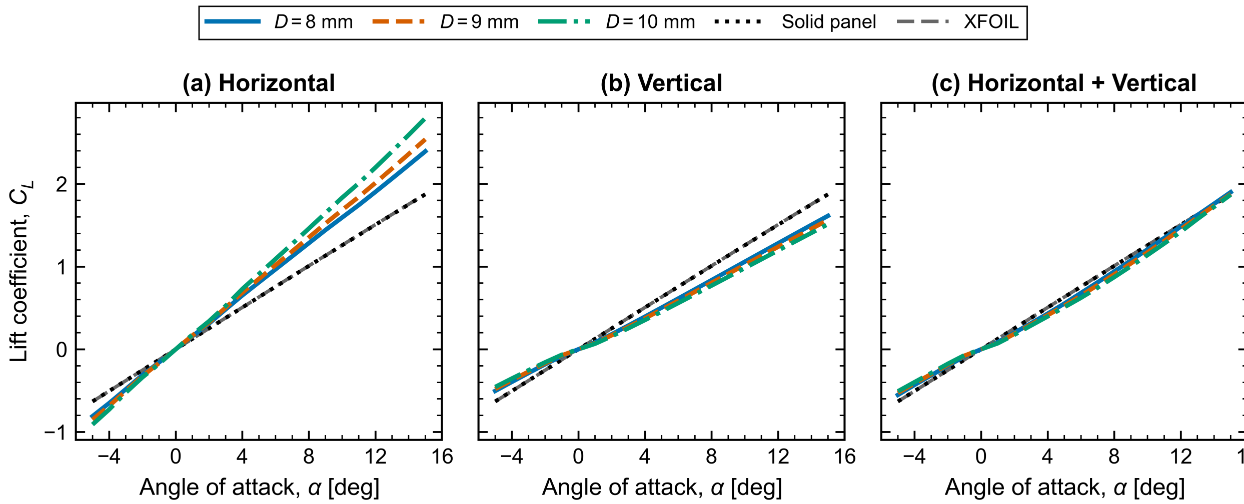


Figure 3: Lift coefficient as a function of angle of attack for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Results are shown for circular-channel diameters $D = 8$, 9 and 10 mm together with the impermeable panel-method and XFOIL reference solutions.

3.2. Effect of porous-channel configuration on lift

Figure 3 compares the variation of lift coefficient with angle of attack for the three porous configurations and channel diameters.

The horizontal configuration produces the largest modification of the lift response. At positive angles of attack, the porous horizontal cases develop a larger lift coefficient than the impermeable reference, and the difference increases progressively with angle of attack. Increasing the channel diameter further strengthens this effect, with the $D = 10$ mm configuration producing the largest lift coefficient over the upper part of the investigated angle-of-attack range.

The same diameter dependence is also apparent at negative angles of attack, where increasing the channel diameter increases the magnitude of the negative lift. The approximately antisymmetric response about the zero-lift region is consistent with the symmetric NACA 0018 geometry and indicates that the horizontal channels primarily modify the aerodynamic loading generated as the angle of attack departs from zero.

The vertical configuration exhibits a distinctly different response. At positive angles of attack, the lift coefficient is generally lower than that of the impermeable airfoil. Furthermore, the three channel diameters remain relatively close to one another throughout most of the sweep. The aerodynamic influence of the vertical configuration is therefore less sensitive to channel diameter than that of the horizontal arrangement.

For the combined horizontal–vertical configuration, the porous solutions remain relatively close to the impermeable reference over a large part of the angle-of-attack range. At positive angles of attack, the lift coefficient is generally slightly lower than the solid-airfoil solution before approaching the reference towards the upper end of the sweep. The three channel diameters also remain closely grouped.

These results demonstrate that the aerodynamic effect of porosity depends strongly on the orientation of the internal channels rather than on channel diameter alone. In particular, increasing diameter does not produce a universal increase in lift. A strong diameter effect is observed for M1, whereas M2 and M3 show considerably weaker sensitivity. This indicates that the pressure difference established between the connected surface locations is a critical factor governing the resulting internal flow and aerodynamic response.

3.3. Effect of porous-channel configuration on pitching moment

The corresponding quarter-chord pitching-moment coefficients are presented in Fig. 4.

The horizontal configuration again produces the strongest departure from the impermeable-airfoil behaviour. As the angle of attack increases, the porous cases develop an increasingly negative pitching moment relative to the solid reference. The magnitude of this change also increases with channel diameter, with the $D = 10$ mm case showing the largest negative pitching moment at high positive angles of attack.

The increase in lift produced by M1 is therefore accompanied by a corresponding modification of the pitching moment. This represents an important aerodynamic trade-off: the configuration that produces the largest lift enhancement also causes the largest change in the longitudinal aerodynamic loading.

The vertical configuration produces substantially smaller changes in pitching moment. The three channel diameters remain closely grouped throughout the angle-of-attack sweep, consistent with the relatively weak diameter sensitivity observed in the corresponding lift results. At positive angles of attack, the porous configurations become slightly more negative than the impermeable reference as the angle of attack increases.

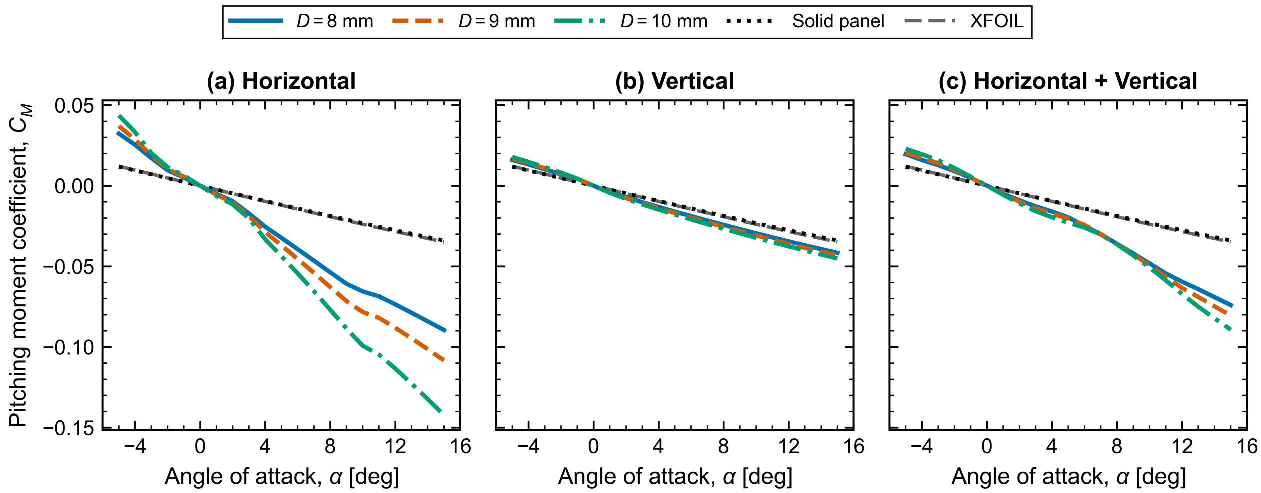


Figure 4: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Results are shown for circular-channel diameters $D = 8$, 9 and 10 mm together with the impermeable panel-method and XFOIL reference solutions.

The combined configuration shows behaviour between that of the horizontal and vertical arrangements. The pitching moment becomes more negative at increasing positive angle of attack, although the departure from the impermeable solution remains smaller than that obtained for the horizontal configuration. Diameter sensitivity also becomes more visible towards the upper end of the angle-of-attack range.

The lift and pitching-moment results therefore show that channel orientation affects not only the magnitude of the aerodynamic force but also its distribution along the chord. The particularly strong changes observed for M1 suggest that the horizontal passages cause a larger redistribution of surface pressure than the vertical passages.

3.4. Flow-field modification induced by the porous channels

To examine the origin of the changes observed in the integrated aerodynamic coefficients, the porous-airfoil pressure and velocity fields are compared with the corresponding impermeable-airfoil solution. The comparison is performed at

$$\alpha = 10^\circ, \qquad D = 9 \text{ mm}. \tag{42}$$

This condition provides a representative positive angle of attack at which the aerodynamic effect of the porous channels is clearly developed.

3.4.1. Pressure-field modification

Figure 5 shows the pressure difference

$$\Delta p_{\text{field}} = p_{\text{porous}} - p_{\text{solid}}, \tag{43}$$

for the three porous configurations.

The pressure changes are concentrated primarily around the channel openings, confirming that the imposed transpiration modifies the local external pressure field. Adjacent regions of positive and negative pressure difference occur around several openings, indicating local redistribution of the aerodynamic loading rather than a uniform pressure shift over the complete surface.

For the horizontal configuration, the pressure differences are predominantly localised around the individual chordwise openings. Distinct pressure increases and reductions are generated on both the upper and lower surfaces. Although these regions remain spatially localised, their cumulative influence modifies the integrated surface loading sufficiently to produce the substantial increase in lift and pitching-moment magnitude observed in Figs. 3 and 4.

The vertical configuration produces a different spatial pattern. In addition to the local pressure changes at individual openings, a comparatively broad pressure modification develops around the leading-edge region. Since each vertical passage connects the lower and upper surfaces, the pressure-driven internal flow directly couples regions subjected to different external aerodynamic loading.

The combined configuration contains features associated with both the horizontal and vertical arrangements. Localised pressure changes are observed at the horizontal-channel openings, while the vertical passages introduce additional upper-to-lower surface interaction. However, the resulting integrated lift response does not equal the strong enhancement obtained for the horizontal model alone. This demonstrates that superposing the two channel arrangements does not produce a simple additive aerodynamic effect.

3.4.2. Velocity-field modification

The corresponding difference in external velocity magnitude is shown in Fig. 6. The plotted quantity is

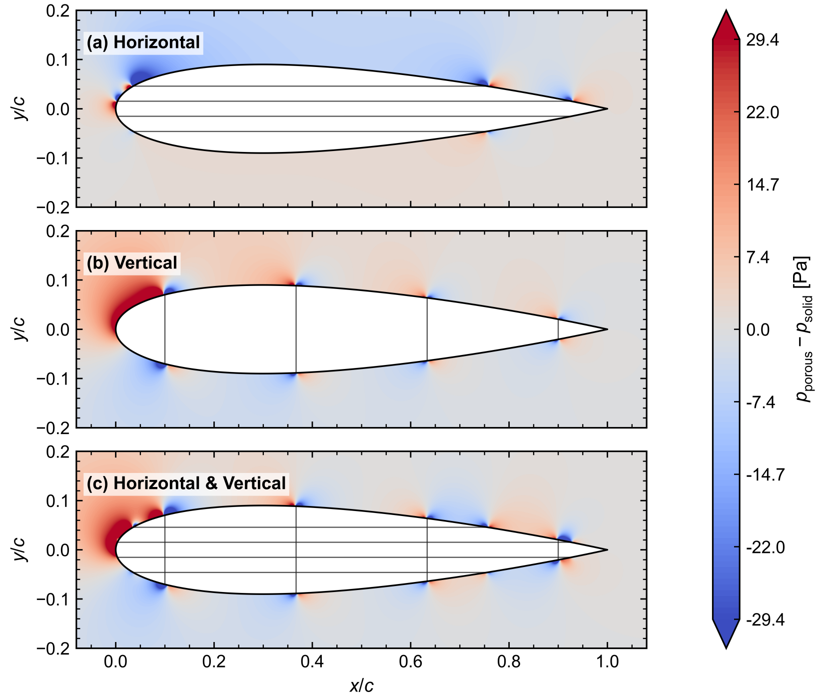


Figure 5: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10^\circ$ and $D = 9$ mm for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Positive values represent an increase in pressure relative to the impermeable airfoil and negative values represent a reduction.

$$\Delta|\boldsymbol{V}| = |\boldsymbol{V}|_{\text{porous}} - |\boldsymbol{V}|_{\text{solid}}. \tag{44}$$

The velocity-field differences show a spatial distribution consistent with the pressure-field modifications. The strongest changes occur close to the channel openings, where the surface-normal transpiration directly alters the local external flow.

The horizontal configuration produces alternating regions of acceleration and deceleration around the chordwise openings. The modifications remain concentrated close to the airfoil surface, although their influence extends into the surrounding external flow. These local changes alter the surface velocity and therefore the pressure coefficient through the Bernoulli relation.

For the vertical configuration, a broader velocity modification is visible around the leading-edge region together with local changes at the remaining channel openings. This behaviour corresponds closely to the pressure redistribution observed in Fig. 5. The resulting modification of the upper- and lower-surface velocities reduces the lift relative to the impermeable case over much of the positive angle-of-attack range.

The combined configuration again exhibits characteristics of both M1 and M2. Velocity changes are present around both the horizontal and vertical opening locations, but their interaction produces a different overall aerodynamic response from either individual configuration.

3.5. Overall comparison of porous configurations

The combined aerodynamic and flow-field results demonstrate that the orientation of the internal channels is the dominant parameter governing the effect of the passive porous system.

Among the configurations investigated, the horizontal model produces the strongest modification of the aerodynamic response. At positive angles of attack, M1 increases the lift coefficient relative to the impermeable reference, with the magnitude of the enhancement increasing with channel diameter. However, this lift enhancement is accompanied by a progressively more negative pitching moment, particularly for the larger channel diameters.

The vertical configuration produces a comparatively weaker dependence on channel diameter and generally reduces the lift coefficient relative to the impermeable airfoil at positive angles of attack. Its effect on pitching moment is also considerably smaller than that of the horizontal configuration.

The combined horizontal–vertical configuration does not simply superpose the aerodynamic behaviour of M1 and M2. Despite containing the horizontal passages responsible for the strongest lift enhancement in M1, the additional vertical passages alter the pressure and velocity redistribution such that the overall lift response remains much closer to the impermeable-airfoil solution.

The pressure- and velocity-difference fields support these integrated aerodynamic trends. The strongest local modifications occur around the channel openings, while the orientation and connectivity of the passages determine how

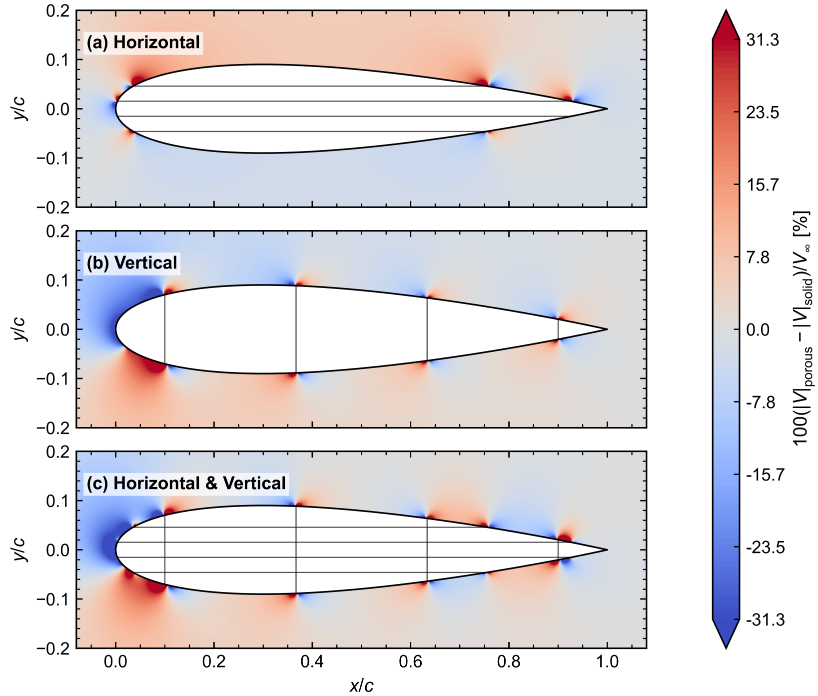


Figure 6: Difference in velocity magnitude between the porous and impermeable-airfoil solutions at $\alpha = 10°$ and $D = 9$ mm for (a) the horizontal model (M1), (b) the vertical model (M2), and (c) the combined horizontal–vertical model (M3). Positive values indicate an increase in velocity magnitude and negative values indicate a reduction relative to the impermeable solution.

these local changes are distributed between the upper and lower airfoil surfaces. Consequently, the aerodynamic performance of the porous airfoil depends not only on channel size but also strongly on the placement and orientation of the internal flow paths.

Overall, the horizontal configuration provides the greatest potential for increasing lift among the three configurations investigated. However, the accompanying pitching-moment change indicates that the selection of a porous-channel arrangement should consider the combined effect on aerodynamic force and moment rather than lift enhancement alone.

4. Installation

The package is available at author resources page at Elsevier (`http://www.elsevier.com/locate/latex`). The class may be moved or copied to a place, usually, `$TEXMF/tex/latex/elsevier/`, or a folder which will be read by LaTeX during document compilation. The TeX file database needs updation after moving/copying class file. Usually, we use commands like `mktexlsr` or `texhash` depending upon the distribution and operating system.

5. Front matter

The author names and affiliations could be formatted in two ways:

(1) Group the authors per affiliation.

(2) Use footnotes to indicate the affiliations.

See the front matter of this document for examples. You are recommended to conform your choice to the journal you are submitting to.

6. Bibliography styles

There are various bibliography styles available. You can select the style of your choice in the preamble of this document. These styles are Elsevier styles based on standard styles like Harvard and Vancouver. Please use BibTeX to generate your bibliography and include DOIs whenever available.

Here are two sample references: [1] [1, 2] [1, 3]

7. Floats

Figures may be included using the command, `\includegraphics` in combination with or without its several options to further control graphic. `\includegraphics` is provided by graphic[s,x].sty which is part of any standard LaTeX distribution. graphicx.sty is loaded by default. LaTeX accepts figures in the postscript format while pdfLaTeX accepts *.pdf, *.mps (metapost), *.jpg and *.png formats. pdfLaTeX does not accept graphic files in the postscript format.

The `table` environment is handy for marking up tabular material. If users want to use multirow.sty, array.sty, etc., to fine control/enhance the tables, they are welcome to load any package of their choice and cas-dc.cls will work in combination with all loaded packages.

Figure 7: The beauty of Munnar, Kerala. (See also Table 1).

Table 1
This is a test caption. This is a test caption. This is a test caption. This is a test caption. Use {table*} instead of {table} if you want a two column spanned table.

Col 1	Col 2	Col 3	Col4
12345	12345	123	12345
12345	12345	123	12345
12345	12345	123	12345
12345	12345	123	12345
12345	12345	123	12345

8. Theorem and theorem like environments

cas-dc.cls provides a few shortcuts to format theorems and theorem-like environments with ease. In all commands the options that are used with the `\newtheorem` command will work exactly in the same manner. cas-dc.cls provides three commands to format theorem or theorem-like environments:

```
\newtheorem{theorem}{Theorem}
\newtheorem{lemma}[theorem]{Lemma}
\newdefinition{rmk}{Remark}
\newproof{pf}{Proof}
\newproof{pot}{Proof of Theorem \ref{thm2}}
```

The `\newtheorem` command formats a theorem in LaTeX's default style with italicized font, bold font for theorem heading and theorem number at the right hand side of the theorem heading. It also optionally accepts an argument which will be printed as an extra heading in parentheses.

```
\begin{theorem}
For system (8), consensus can be achieved with
$\|T_{\omega z}$ ...
\begin{eqnarray}\label{10}
....
\end{eqnarray}
\end{theorem}
```

Theorem 1. *For system (8), consensus can be achieved with* $\|T_{\omega z}$ ...

$$.... \tag{45}$$

The `\newdefinition` command is the same in all respects as its `\newtheorem` counterpart except that the font shape is roman instead of italic. Both `\newdefinition` and `\newtheorem` commands automatically define counters for the environments defined.

The `\newproof` command defines proof environments with upright font shape. No counters are defined.

9. Enumerated and Itemized Lists

cas-dc.cls provides an extended list processing macros which makes the usage a bit more user friendly than the default LaTeX list macros. With an optional argument to the `\begin{enumerate}` command, you can change the list counter type and its attributes.

```
\begin{enumerate}[1.]
\item The enumerate environment starts with an optional
argument `1.', so that the item counter will be suffixed
by a period.
\item You can use `a)' for alphabetical counter and '(i)'
for roman counter.
\begin{enumerate}[a)]
\item Another level of list with alphabetical counter.
\item One more item before we start another.
\item One more item before we start another.
\item One more item before we start another.
\item One more item before we start another.
```

Further, the enhanced list environment allows one to prefix a string like 'step' to all the item numbers.

```
\begin{enumerate}[Step 1.]
\item This is the first step of the example list.
\item Obviously this is the second step.
\item The final step to wind up this example.
\end{enumerate}
```

10. Cross-references

In electronic publications, articles may be internally hyperlinked. Hyperlinks are generated from proper cross-references in the article. For example, the words Fig. 1 will never be more than simple text, whereas the proper cross-reference `\ref{tiger}` may be turned into a hyperlink to the figure itself: Fig. 1. In the same way, the words Ref. [1] will fail to turn into a hyperlink; the proper cross-reference is `\cite{Knuth96}`. Cross-referencing is possible in LaTeX for sections, subsections, formulae, figures, tables, and literature references.

11. Bibliography

Two bibliographic style files (`*.bst`) are provided — model1-num-names.bst and model2-names.bst — the first one can be used for the numbered scheme. This can also be used for the numbered with new options of natbib.sty.

Figure 8: The beauty of Munnar, Kerala. (See also Table 1).

The second one is for the author year scheme. When you use model2-names.bst, the citation commands will be like `\citep`, `\citet`, `\citealt` etc. However when you use model1-num-names.bst, you may use only `\cite` command.

`thebibliography` environment. Each reference is a `\bibitem` and each `\bibitem` is identified by a label, by which it can be cited in the text:

In connection with cross-referencing and possible future hyperlinking it is not a good idea to collect more that one literature item in one `\bibitem`. The so-called Harvard or author-year style of referencing is enabled by the LaTeX package natbib. With this package the literature can be cited as follows:

- Parenthetical: `\citep{WB96}` produces (Wettig & Brown, 1996).
- Textual: `\citet{ESG96}` produces Elson et al. (1996).
- An affix and part of a reference: `\citep[e.g.][Ch. 2]{Gea97}` produces (e.g. Governato et al., 1997, Ch. 2).

In the numbered scheme of citation, `\cite{<label>}` is used, since `\citep` or `\citet` has no relevance in the numbered scheme. natbib package is loaded by cas-dc with `numbers` as default option. You can change this to author-year or harvard scheme by adding option `authoryear` in the class loading command. If you want to use more options of the natbib package, you can do so with the `\biboptions` command. For details of various options of the natbib package, please take a look at the natbib documentation, which is part of any standard LaTeX installation.

A. My Appendix

Appendix sections are coded under `\appendix`.

`\printcredits` command is used after appendix sections to list author credit taxonomy contribution roles tagged using `\credit` in frontmatter.

CRediT authorship contribution statement

J.K. Krishnan: Conceptualization of this study, Methodology, Software. **William J. Hansen:** Data curation, Writing - Original draft preparation.

References

[1] Fortunato, S., 2010. Community detection in graphs. Phys. Rep.-Rev. Sec. Phys. Lett. 486, 75–174.

[2] Newman, M.E.J., Girvan, M., 2004. Finding and evaluating community structure in networks. Phys. Rev. E. 69, 026113.

[3] Vehlow, C., Reinhardt, T., Weiskopf, D., 2013. Visualizing fuzzy overlapping communities in networks. IEEE Trans. Vis. Comput. Graph. 19, 2486–2495.

Author biography without author photo. Author biography.

Author biography with author photo. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography.

Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography.

Author biography with author photo. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography. Author biography.